Enthusiast Course

PHASE : SPEED & ACCURACY TEST

TARGET : PRE - MEDICAL 2022

Test Type :**PRACTICE TEST** Test Pattern :**NEET (UG)**

TEST DATE :

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	3	4	3	1	1	2	4	2	4	2	3	2	4	2	3	2	1	1	4	1	4	3	3	1	2	3	3	3	4

Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	1	3	1	2	1	1	3	3	4	2	1	3	4	4	3

HINT – SHEET

1. Ans (4)

Huygen's construction of wavefront does not apply to the origin of spectra which is explained by quantum theory.

2. Ans (3)

Phase difference$=\dfrac{2\pi x}{\lambda}$ or $\lambda = \dfrac{2\pi x}{\Phi}$

$\lambda = 2\pi \dfrac{(2.1X10^{-6})}{7.692\pi} = 5460\ X10^{-10}$ m

3. Ans (4)

By Malus law, $I = I_0\cos^2\theta$

(I_0 = intensity of emergent polarized light)

Where $\theta = 60°$, I =?

$I = I_0$ x $\cos^2 60°$

I_0= intensity passed through polarizer)

$= I_0$ x $(1/2)^2 = I_0/4$

4. Ans (3)

Fringe width β= 2 mm

λ= 6000 Å

μ=1.33. $\beta' = \dfrac{\beta}{\mu}$

=1.5 mm

5. Ans (1)

All electromagnetic waves are polarized or polarizable in nature (light is an Electromagnetic wave and that's why my sunglasses are able to block part of it by allowing light of only certain polarization through them). Sound wave cannot be polarized due to its longitudinal wave nature.

6. Ans (1)

P_1 is at central maxima so $I_1 = 4I_o$

For P_2,

$$\phi = \left(\frac{2\pi}{\beta}\right)\left(\frac{\beta}{4}\right) = \frac{\pi}{2}$$

$$I_2 = I_o + I_o + \sqrt{I_o I_o}\cos\left(\frac{\pi}{2}\right) = 2I_o$$

$$\frac{I_1}{I_2} = 2$$

7. Ans (2)

Path difference on circles around 'P' is same. So, the fringes obtained on the screen in the given condition will be concentric circles.

8. Ans (4)

For the secondary minima $d\sin\theta = n\lambda$

$\sin\theta = n\lambda/d$

For second minima $n = 2$

$\sin\theta = \frac{2\lambda}{d} = \tan\theta_1 = \frac{x_1}{D}$

For fourth minima $n = 4$

$\sin\theta_2 = \frac{4\lambda}{d} = \frac{x_2}{D}$

$x_2 - x_1 = \frac{4\lambda}{d} - \frac{2\lambda}{d} = \frac{2\lambda}{d} = 6 - 3 = 3\text{cm}$

Width of the central max = 3 cm

9. Ans (2)

Since there is no shift in central maxima, therefore path difference introduced by the two sheets are equal

i.e., $(\mu_1 - 1)\, t_1 = (\mu_2 - 1)\, t_2$

where μ_1 and μ_2 are refraction index

i.e., $\frac{t_1}{t_2} = \frac{(\mu_2 - 1)}{(\mu_1 - 1)} = \frac{1.5 - 1}{1.25 - 1} = \frac{0.5}{0.25} = 2$

10. Ans (4)

Position of dark fringe

$d\sin\theta = (2n - 1)\frac{\lambda}{2}$

$n = 1$

$\sin\theta = \frac{\lambda}{2d}$

$\sin\theta = \frac{5460 \times 10^{-10}}{2 \times 10^{-4}}$

$\sin\theta \approx \theta = 2730 \times 10^{-6} \times \frac{180}{\pi}$

$\theta = 1.56496 \approx 0.16$

11. Ans (2)

Given λ= 600 nm = 600 x 10^{-9} m = 6 x 10^{-7} m

$a\frac{x}{D} = n\lambda$

$a = n\lambda\frac{D}{x}$

$a = 2 \times 6 \times 10^{-7} \times \frac{0.8}{1.5 \times 10^{-3}}$

$a = 6.4 \times 10^{-4}\text{m}$

12. Ans (3)

$x = \frac{\beta}{\lambda}(\mu - 1)t$

$\Rightarrow 4\beta = \frac{\beta}{\lambda}(\mu - 1)t$

$\Rightarrow t = \frac{4\lambda}{(\mu - 1)}$

13. Ans (2)

As given,

$\frac{D}{d} \times 3 \times 6 \times 10^{-7} = \frac{D}{d} \times 4 \times \lambda_2$

$\lambda_2 = \frac{3 \times 6 \times 10^{-7}}{4}$

$\lambda_2 = 4.5 \times 10^{-7}\text{m} = 4500\text{Å}$

14. Ans (4)

Intensity of dark and bright fringes is given by

$I_{dark} = (\sqrt{I_1} - \sqrt{I_2})^2$

$I_{bright} = (\sqrt{I_1} + \sqrt{I_2})^2$

When there is no glass plate

$I_1 = I_2 = I_0$

$I_{dark} = 0$ and

$I_{bright} = 4I_0$

But when glass plate is introduced in the path one of its slits,

$I_2 = \frac{I_0}{2}$

So

$I_{dark} \neq 0$ increases and I_{bright} decreases.

Fringe width remains same.

Hence, Bright will become less bright and dark will become less dark.

15. Ans (2)

For maxima

$\Delta x = d\sin\theta = n\lambda$

$\therefore \quad \sin\theta = \frac{n\lambda}{d} = \frac{n \times 2000}{7000}$

since $\sin\theta$ cannot be greater than 1.

$\therefore$ *n=0,1,2,3 only.*

$\therefore$ 3 maxima on the both side of screen and 1 central maxima will be obtained.

$\therefore$ *Total maxima 7*

16. **Ans (3)**

$$\frac{I_{max}}{I_{min}} = \left[\frac{a_1 + a_2}{a_1 - a_2}\right]^2$$

$$\frac{9}{1} = \left[\frac{a_1 + a_2}{a_1 - a_2}\right]^2$$

$$\frac{a_1 + a_2}{a_1 - a_2} = \frac{3}{1}$$

By solving this

$$a_1 + a_2 = 3a_1 - 3a_2$$

$$4a_2 = 2a_1$$

$$\frac{a_1}{a_2} = \frac{2}{1}$$

17. **Ans (2)**

For first minima

$$\sin\theta = \frac{\lambda}{a} = \frac{5000 \times 10^{-10}}{1 \times 10^{-6}}$$

$\theta = 30^o$

so the angular width of the central maxima is

$2\theta = 60^o$

18. **Ans (1)**

At x = 0 path difference is 4λ i.e., fifth bright fringe will be obtained at x = 0 and at x = ∞, path difference is 0, hence first or central bright fringe will be obtained at x = ∞, therefore between 1st and 5th bright fringe, the number of fringes will be 3.

19. **Ans (1)**

Position of fringe from central maxima

$$y_1 = \frac{n\lambda_1 D}{d}$$

Given, n = 10

$$y_1 = \frac{10\lambda_1 D}{d} \quad \ldots \text{(i)}$$

For second source

$$y_2 = \frac{5\lambda_2 D}{d} \quad \ldots \text{(ii)}$$

$$\frac{y_1}{y_2} = \frac{2\lambda_1}{\lambda_2}$$

20. **Ans (4)**

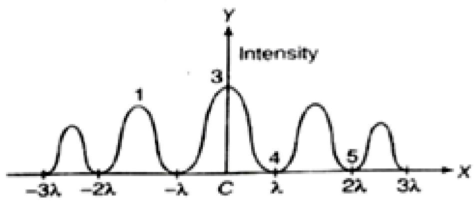
In diffraction of light at a single slit, the pattern obtained consists of a central bright band having alternate dark and bright bands of decreasing intensity on both sides. For n^{th} minima (secondary)

$$\sin\theta_n = \frac{n\lambda}{a}$$

for n^{th} maxima (secondary)

$$\sin\theta_n = (2n + 1)\frac{\lambda}{2a}$$

Thus, point 5 corresponds to the path difference of extreme rays equal to 2λ



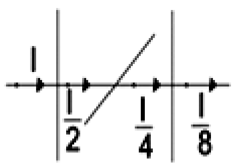
21. **Ans (1)**

Since fringe width $\beta = \dfrac{\lambda D}{d}$

When screen is moved away, D increases. Therefore, width of the fringes increases but the angular separation (λ/d) remains the same.

22. **Ans (4)**

Transformation axis A and B are parallel if transformation of Axis of C is making angle θ with A, it will also make angle θ with B



$$\frac{1}{2}\cos^2\theta\cos^2\theta = \frac{1}{8}$$

$$\cos^4\theta = \frac{1}{4} \Rightarrow \cos\theta = \frac{1}{\sqrt{2}}$$

$\theta = 45^o$

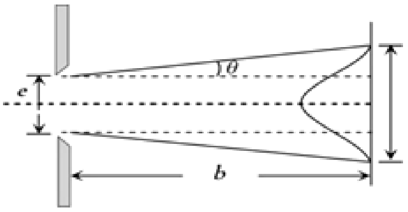
23. Ans (3)

The direction in which the first minima occur is θ (say).

The direction in which the first minima occurs is θ.

Then $e\sin\theta = \lambda$ or $e\theta = \lambda$

$\theta = \frac{\lambda}{e}$ $\because$ $\sin\theta = \theta$ when θ is small.



Width of the central maximum

$= 2b\theta + e = \frac{2b\lambda}{e} + e$

24. Ans (3)

In the double slit experiment

$I_{net} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\emptyset$

When $I_1 = I_2$ and $\cos\emptyset = +1$

We get $I_{max} = 4I_1$

Maximum intensity in the double slit interference = 4 (maximum intensity of single slit experiment)

25. Ans (1)

As $\beta = \frac{\lambda D}{d}$ and $\lambda_b < \lambda_y$

Fringe width β will decrease

26. Ans (2)

Let I_0 be the intensity of incident light. As the light coming from sodium Iamp is unpolarized, so the intensity of light emerging from the first polaroid is

$I_1 = \frac{I_0}{2}$

If θ is the angle between two Polaroids, then the intensity of light emerging from the second polaroid is

$I_2 = I_1\cos^2\theta = \frac{I_0}{2}\cos^2\theta$

But $I_0 = 100\%$ (given)

$\therefore I_2 = (50\%)\cos^2\theta$

Since θ varies from 90^0 to 0^0, so the intensity of the out coming light can be varied from 0% to 50%.

27. Ans (3)

For 2^{nd} minima

$d\sin\theta = 2\lambda$

$d\sin 60^o = 2\lambda$

$d = \frac{4\lambda}{\sqrt{3}}$

So, for first minima

$d\sin\theta_1 = \lambda$

$\sin\theta_1 = \frac{\lambda}{d} = \frac{\sqrt{3}}{4}$

$\theta_1 = 25.65^\circ \approx 25^\circ$

28. Ans (3)

For first minimum,

$\sin 30^o = \frac{\lambda}{a}$

$\frac{1}{2} = \frac{\lambda}{a}$

$a = 2\lambda$

Now, for 1^{st} secondary maxima

$\sin\theta = \frac{3\lambda}{2a}$

$\sin\theta = \frac{3\lambda}{2\times 2\lambda}$

$\sin\theta = \frac{3}{4}$

$\theta = \sin^{-1}\left(\frac{3}{4}\right)$

29. Ans (3)

Interference is on the principle of conservation of energy. In interference, energy is redistributed.

30. Ans (4)

$I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2$ and α

$= 0, \frac{3\pi}{2}, \frac{5\pi}{2}$ …… for maxima

The ratio of intensities of successive maxima is

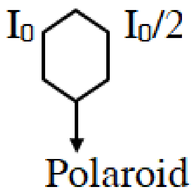
$I_n = I_0\left(\frac{2}{(2n+1)\pi}\right)^2$

$1 : \left(\frac{2}{3\pi}\right)^2 : \left(\frac{2}{5\pi}\right)^2$ or $1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2}$

31. Ans (1)

According to Brewster's law, polarizing angle depends on wavelength and is different for different colours.

32. Ans (3)



$I_0 = A'^2$ ....(1)

$\frac{I_0}{2} = A^2$ ....(2)

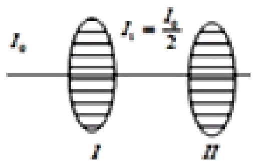
(1) ÷ (2)

$$\frac{I_0}{I_0/2} = \frac{A'^2}{A^2}$$

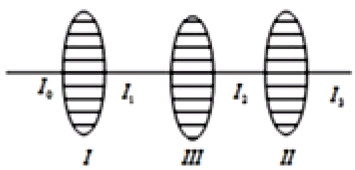
$$\boxed{(A') = 2A^2 = I_0}$$

Due to polarization, intensity of unpolarized light will reduce to half of its value

33. Ans (1)



If no light comes from 2^{nd}polaroid , i.e. angle between them = 90



If 3^{rd}polaroid has polarization axis at angle θ

$I_2 = I_1 \cos^2\theta$

$I_2 = \frac{I_0}{2}\cos^2\theta$

As polarizing axis of 1^{st} and 2^{nd} are mutually perpendicular

Hence angle between polarizing axis of 2^{nd} and 3^{rd} is 90^o - θ

$I_3 = I_2 \cos^2(90^o - \theta)$

$\Rightarrow I_3 = \frac{I_0}{2}\cos^2\theta\sin^2\theta$

$\Rightarrow I_3 = \frac{I_0}{2} \times \frac{4}{4}\cos^2\theta\sin^2\theta$

$\Rightarrow I_3 = \frac{I_0}{8}(2\cos\theta\sin\theta)^2$

$\Rightarrow I_3 = \frac{I_0}{8}\sin^2(2\theta)^2$

34. Ans (2)

Resolving power

$$\frac{1}{d\theta} = \frac{D}{1.22\lambda}$$

$$= \frac{0.61}{1.22 \times 5000 \times 10^{-10}} = 10^6$$

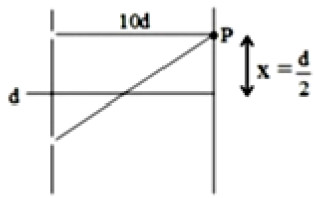
35. Ans (1)

$S_1P - S_2P = \frac{3\lambda}{2}$

or $\sqrt{4\lambda^2 + x^2} - x = \frac{3\lambda}{2}$

on solving $x = \frac{7\lambda}{12}$

36. Ans (1)



Path difference at point P is

$\Delta x = \frac{dx}{D} = \frac{d^2}{2D}$

$\Delta x = \frac{(5\lambda)^2}{2 \times 10d}$

$\Delta x = \frac{(5\lambda)^2}{2 \times 10 \times 5\lambda} = \frac{\lambda}{4}$

Therefore, phase difference is

$\Delta\phi = \frac{2\pi}{\lambda} \times \Delta x = \frac{\pi}{2}$

If I is the intensity due to each slit, then the maximum intensity is-

$I_0 = 4I$

$I = \frac{I_0}{4}$

Therefore, intensity at point P is

$I_{net} = I + I + 2\sqrt{I}\sqrt{I}\cos\frac{\pi}{2}$

$I_{net} = 2I = \frac{I_0}{2}$

37. **Ans (3)**

Minimum angular separation between two stars whose image is just resolved by telescope is

$$\Delta\theta = \frac{1.22\lambda}{a}$$

$$\therefore \quad \Delta\theta = \frac{1.22 \times 5000 \times 10^{-1} \times 10^{-10}}{2}$$

$$\Delta\theta = 3050 \times 10^{-10} = 3.05 \times 10^{-7}\text{rad}$$

$$\Delta\theta \approx 0.31 \times 10^{-6} \quad \text{rad}$$

38. **Ans (3)**

By placing a sheet optical path length changes by (μ-1)t using that justify that by placing a sheet fringe width remains same only entire wave pattern shifts by a distance $\Delta y = \frac{(\mu - 1)D}{d}$.

39. **Ans (4)**

$$\Delta\emptyset = \frac{2\pi}{\lambda} \times \frac{\lambda}{8} = \pi/4$$

$$I = I_0\cos^2\frac{\pi}{8} = I_0\left(\frac{1 + \cos\frac{\pi}{4}}{2}\right)$$

$$= I_0\left(\frac{1 + 1/\sqrt{2}}{2}\right) = 0.85I_0$$

40. **Ans (2)**

$$(\mu - 1)\,t = n\lambda$$

$$\text{or } t = \frac{n\lambda}{\mu - 1} = \frac{5 \times 5000 \times 10^{-10}}{1.5 - 1}$$

$$= \frac{5 \times 5000 \times 10^{-10}}{\frac{1}{2}} = 10 \times 5000 \times 10^{-10}$$

$$= 5 \times 10^{-6}\text{ m} = 5\mu\text{m}$$

41. **Ans (1)**

For reflected system of the film, the condition for the maxima is $2\mu t\cos r = (2n - 1)\frac{\lambda}{2}$. While the maxima for transmitted system of film is the $2\mu t\cos r = n\lambda$.

42. **Ans (3)**

$$y_1 = \frac{n_1\lambda_1 D}{d} \text{ for bright fringes}$$

$$y_2 = \frac{n_2\lambda_2 D}{d} \text{ for bright fringes}$$

To coincide

$$n_1\lambda_1 = n_2\lambda_2$$

$$n_1 \times 650 = n_2 \times 520$$

$$\frac{n_1}{n_2} = \frac{4}{5}$$

For minimum value of n_1 and n_2

$$n_1 = 4 \ \& \ n_2 = 5$$

$$(y_1)_{min} = 4 \times 650 \times 10^{-9}\text{m} \times \frac{150\text{m}}{0.5 \times 10^{-3}\text{m}}$$

$$= 7800 \times 10^{-6}\text{ m} = 7.8\text{ mm}$$

43. **Ans (4)**

By polarization due to reflection at polarizing angle reflected light is completely polarized and the intensity is less than the average intensity $\frac{I_0}{2}$

44. **Ans (4)**

The fringe width in young's double slit experiment

$$\beta = \frac{\lambda D}{d}$$

where, λ = wavelength of the light used.

D is the distance between slit and screen.

d is the distance between the slit.

$$\beta' = \frac{\lambda(2D)}{d/2} = \frac{4\lambda D}{d} = 4\beta$$

45. Ans (3)

Distance of second minima from central point is $Y = \frac{d}{2}$

We know that minima is given by

$$d \sin\theta = \left(n - \frac{1}{2}\right)\lambda$$

$$\frac{dY}{D} = \left(2 - \frac{1}{2}\right)\lambda$$

$$\frac{d}{2} = \frac{D}{d}\left(\frac{2 \times 2 - 1}{2}\right)\lambda$$

$$\therefore \quad \frac{d}{2} = \frac{D}{d}\frac{3}{2}\lambda$$

$$\therefore \quad \lambda = \frac{d^2}{3D}$$